

Discrete optimization

Relaxation

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Optimization: principles and algorithms



ÉCOLE POLYTECHNIQUE
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Relaxation

We know how to solve linear optimization problems.

We do not know how to solve discrete optimization problems.

But if we forget about the integrality constraints, we obtain a linear optimization problem.

It is called a relaxation, and happens to be very useful.

Relaxation

Remove by hand the y and replace z by $0 \leq z \leq 1$.

Relaxation

$$\min_{x \in \mathbb{R}^{n_x}, y \in \mathbb{Z}^{n_y}, z \in \mathbb{N}^{n_z}} f(x, y, z)$$

subject to

$$g(x, y, z) \leq 0$$

$$h(x, y, z) = 0$$

$$y \in \mathbb{Z}^{n_y}$$

$$z \in \{0, 1\}^{n_z}$$

where

- ▶ $f : \mathbb{R}^{n_x} \times \mathbb{R}^{n_y} \times \mathbb{R}^{n_z} \rightarrow \mathbb{R},$
- ▶ $g : \mathbb{R}^{n_x} \times \mathbb{R}^{n_y} \times \mathbb{R}^{n_z} \rightarrow \mathbb{R}^m,$
- ▶ $h : \mathbb{R}^{n_x} \times \mathbb{R}^{n_y} \times \mathbb{R}^{n_z} \rightarrow \mathbb{R}^p.$

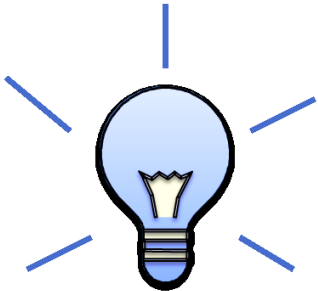
Lower bound

- ▶ Discrete optimization P : optimal solution: (x^*, y^*, z^*) .
- ▶ Relaxation $R(P)$: optimal solution: (x_R^*, y_R^*, z_R^*) .

$$f(x_R^*, y_R^*, z_R^*) \leq f(x^*, y^*, z^*).$$

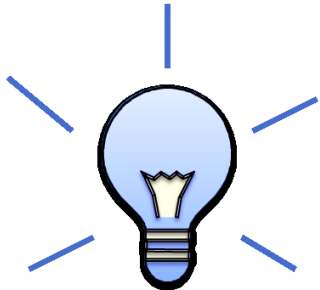
Proof: the integer solution (x^, y^*, z^*) verifies the constraints of the relaxation.
Mention that it is valid only for global minima.*

Mixed Integer Linear Problems



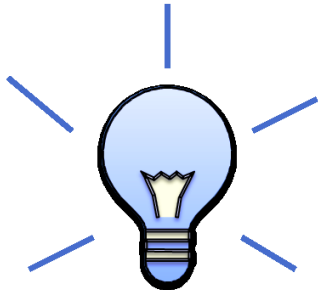
- Consider the relaxation $R(P)$.

Mixed Integer Linear Problems



- ▶ Consider the relaxation $R(P)$.
- ▶ Calculate (x_R^*, y_R^*, z_R^*) using the simplex algorithm.

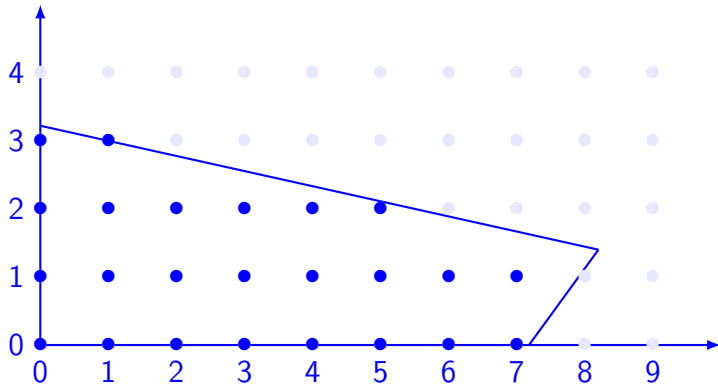
Mixed Integer Linear Problems



- ▶ Consider the relaxation $R(P)$.
- ▶ Calculate (x_R^*, y_R^*, z_R^*) using the simplex algorithm.
- ▶ Round the solution to the nearest integer values.

An example

$$\begin{aligned} & \min_{x \in \mathbb{N}^2} -3x_1 - 13x_2 \\ & \text{subject to} \\ & 2x_1 + 9x_2 \leq 29 \\ & 11x_1 - 8x_2 \leq 79. \end{aligned}$$



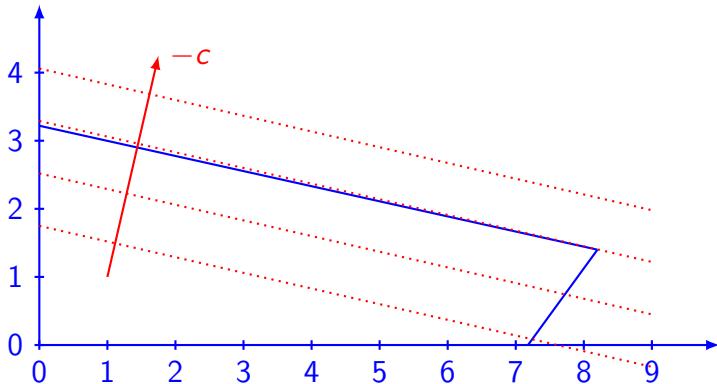
Solving the relaxation

$$\min_{x \in \mathbb{N}^2} -3x_1 - 13x_2$$

subject to

$$2x_1 + 9x_2 \leq 29$$

$$11x_1 - 8x_2 \leq 79.$$



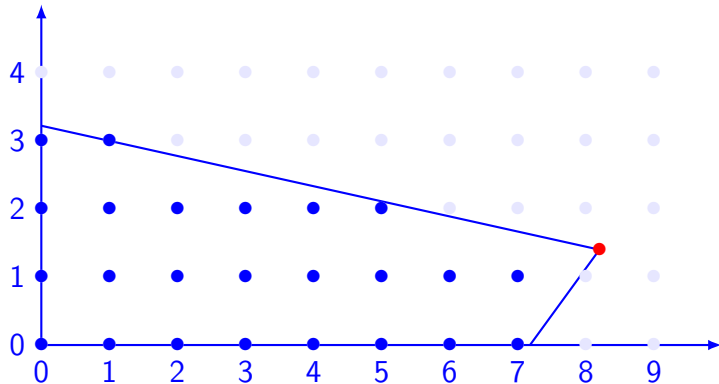
Rounding

Show that rounding always produce an infeasible point

Moreover, the optimal solution $(1, 3)$ is far from the solution of the relaxation.

Rounding the solution

$$\begin{aligned} \min_{x \in \mathbb{N}^2} & -3x_1 - 13x_2 \\ \text{subject to} & \\ & 2x_1 + 9x_2 \leq 29 \\ & 11x_1 - 8x_2 \leq 79. \end{aligned}$$



Summary

The relaxation problem is a discrete optimization problem where the integrality constraints are ignored.

In the context of MILP, it is a linear optimization problem, that can be solved using the simplex algorithm.

Rounding the fractional solution does not provide the solution of the discrete problem.

But solving the relaxation provides a lower bound on the optimal value of the discrete optimization problem.

And this will prove to be extremely useful.